

Simulations of Quantum Approximate Optimization Algorithm on HPC-QC Integrated Systems

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Abstract—The Quantum Approximate Optimization Algorithm (QAOA) has emerged as a promising tool for accelerating optimization processes in the Noisy Intermediate-Scale Quantum (NISQ) era. Compared to classical methods, QAOA efficiently solves optimization problems, often formulated as Quadratic Unconstrained Binary Optimization (QUBO) problems. Classical quantum simulators are crucial for evaluating quantum algorithms due to limited quantum resources. However, QAOA's performance can vary with different simulation methods. This study analyzes QAOA's performance using various quantum simulators (e.g., `density_matrix`, `statevector`, and `matrix_product_state`) and demonstrates the benefits of HPC-QC integrated systems in solving QUBO problems on an active learning workflow. By simulating QAOA on dense, large-matrix QUBO problems, we evaluate accuracy and problem-solving time. We also assess QAOA's performance on local computers and HPC-QC integrated systems, using Oak Ridge Leadership Computing Facility (OLCF)'s Frontier supercomputer with local Qiskit Aer and remote IBM Quantum simulators.

Index Terms—Quantum approximate optimization algorithm, quantum simulator, HPC-QC integrated system, performance analysis, quadratic unconstrained binary optimization

I. INTRODUCTION

Combinatorial optimization problems can be formulated as quadratic unconstrained binary optimization (QUBO) problems [1]. In the noisy intermediate-scale quantum (NISQ) era, quantum computing (QC) shows promise for efficiently addressing these QUBO problems using the quantum approximate optimization algorithm (QAOA). QAOA iteratively updates its variational parameters by leveraging both classical and quantum computing resources [2], [3]. Given the limited resources of real quantum devices, QAOA is often executed with quantum simulators. However, the performance of QAOA can vary depending on the quantum simulation methods employed, due to their distinct operational mechanisms. Furthermore, performance can be enhanced by using HPC-QC integrated systems.

In this work, we simulate QAOA with various quantum simulators to analyze performance. We first generate QUBO problems characterized by dense and large matrices to represent real-world optimization challenges, such as meta-material design. These problems are then solved using QAOA with different quantum simulation methods to evaluate their accuracy and problem-solving time. Additionally, we compare

the performance of QAOA when running on a local computer versus an HPC-QC integrated system.

II. METHODOLOGY

A. QUBO Generation: Given that QUBOs are represented by upper-triangular matrices [4], [5], we generate matrices where elements are confined to the upper-triangular region which are filled with random numbers (from -1 to 1). The size of these QUBO matrices is determined by a bitstring length, and these matrices can be dense and large when the problem size is large. We generate QUBO problems ranging in size from 4 to 32, and analyze the performance of QAOA with various simulation methods by solving given QUBOs.

B. Quantum Simulation Methods: We utilize conventional packages and resources for quantum algorithms and simulators. To generate quantum circuits for QAOA, we use Qiskit (v0.41.0) and Qiskit-optimization (v0.4.0). Additionally, we employ Qiskit for the Basic Aer (QASM) simulator and Qiskit-aer (v0.11.2) for the Aer simulator, using various methods such as `density_matrix`, `statevector`, `matrix_product_state`, and `automatic` [6]. We can execute QAOA jobs on local simulators or submit them to the OLCF Frontier supercomputer, which is loosely integrated with IBM Quantum simulators (IBM/qasm, IBM/statevector, and IBM/mps). Subsequently, the submitted jobs run on the HPC-QC integrated systems. Note that these IBM Quantum simulators were decommissioned in May 2024.

C. HPC-QC Integrated Systems: The Oak Ridge Leadership Computing Facility (OLCF), such as Frontier (9,408 compute nodes, where each node has 64-core AMD "Optimized 3rd Gen EPYC" CPU and 4 AMD Instinct MI250X GPUs with 512 GB of DDR4 memory), is loosely integrated with Quantum Computing User Program resources [7]. These integrated systems allow us to submit hybrid jobs combining classical and quantum computations to both HPC infrastructure and cloud-based quantum devices or simulators. This setup enhances capabilities for quantum simulation and boosts performance for tasks such as QAOA.

III. RESULTS

We first analyze the performance of QAOA executed on a local computer by measuring accuracy (= obtained

solution/optimum solution) and problem-solving time (= measured time from the job submission to obtain a solution). QAOA using the *density_matrix*, *statevector*, and *matrix_product_state* methods yields high accuracy. However, they cannot handle large problems due to their memory-intensive characteristics [8]. The *automatic* method exhibits reliable accuracy, but the accuracy is low for large problems. The cloud-based simulators generally show higher accuracy than local simulators, but they also cannot solve large problems (size > 28). Overall, the accuracy is high for small QUBO problems (size ≤ 10), but it becomes less accurate as the problem size increases (Fig. 1A). The problem-solving time exponentially increases as the problem size increases for all simulators (Fig. 1B). Note that the Aer simulator with *automatic* method and cloud-based simulator (IBM/qasm) exhibit better capabilities in handling large problems, but they also have limitations. For example, solving time is ~30,000 s for a problem size of 30 with the Aer/automatic method.

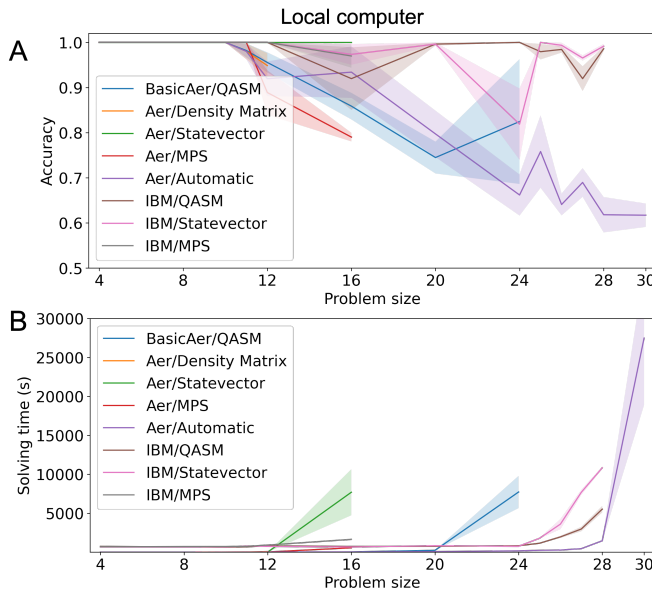


Fig. 1: (A) Accuracy, and (B) solving time of QAOA for QUBO problems executed or submitted on a local computer.

Utilizing integrated HPC-QC systems, such as Frontier with IBM Quantum simulators, enhances the performance of QAOA. As depicted in Figure 2A, a comparative analysis of QAOA performance on standalone HPC versus HPC-QC integrated systems reveals similar results since the operational mechanisms of quantum simulators are the same. However, problem-solving time can be greatly reduced by employing Frontier, especially for large problems (Figs. 1B and 2B). Notably, the HPC-QC system allows for tackling large problem sets effectively due to higher memory capacity.

IV. CONCLUSION

The results highlight the optimal simulator for QAOA, and demonstrate the HPC-QC integrated system can significantly enhance the performance of QAOA. Additionally, We further

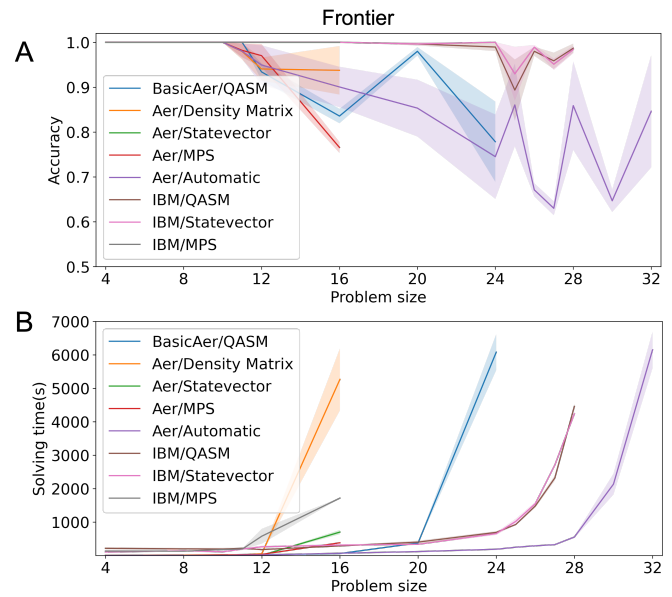


Fig. 2: A) Accuracy, and (B) solving time of QAOA for QUBO problems executed or submitted on an HPC-QC integrated system (Frontier).

develop an optimization algorithm using QAOA with the most efficient simulator to solve QUBO surrogates for real-world problems, such as photonic structure design. This indicates the potential of utilizing QAOA in material science.

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